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Edited By Christopher Dupont

Abstract

Decaying barrels on the seafloor linked to DDT contamination have raised concerns about the public health implications of decades old industrial waste dumped off the coast of Los Angeles. To explore their contents, we collected sediment cores perpendicular to five deep-sea barrels. The concentration of DDT and its breakdown products were highly elevated relative to control sites yet did not vary with distance from the barrels, suggesting that they were not associated with the contamination. Sediment cores collected through white halos surrounding three barrels were enriched in calcite and had elevated pH. The associated microbial communities were low diversity and dominated by alkaliphilic bacteria with metagenome-assembled genomes adapted to high pH. A solid concretion sampled between a white halo and barrel was composed of brucite, a magnesium hydroxide mineral that forms at high pH. Based on these findings, we postulate that leakage of containerized alkaline waste triggered the formation of mineral concretions that are slowly dissolving and raising the pH of the surrounding sediment pore water. This selects for taxa adapted to extreme alkalinity and drives the precipitation of “anthropogenic” carbonates forming white halos, which serve as a visual identifier of barrels that contained alkaline waste. Remarkably, containerized alkaline waste discarded >50 years ago represents a persistent pollutant creating localized mineral formations and microbial communities that resemble those observed at some hydrothermal systems. These formations were observed at one-third of the visually identified barrels in the San Pedro Basin and have unforeseen, long-term consequences for benthic communities in the region.

Keywords: DDT, containerized waste, San Pedro Basin, microbial community, alkalinity, brucite

Significance Statement

Corroding metal barrels detected on the sea floor off Los Angeles have been linked to DDT contamination, yet no connections have been made between these barrels and their contents. Here, we provide evidence that barrels surrounded by white halos and solid concretions contained alkaline waste. This is supported by elevated pore-water pH, the mineral content of the white halos and concretions, and the dominance of alkaliphilic bacteria in the sediment. We show that alkaline waste pollution continues to alter microbial sediment communities more than 50 years after initial disposal and can be considered a persistent pollutant that selects for highly specialized microbial communities related to those observed in hydrothermal seeps and other hyperalkaline habitats.

Introduction

The world's oceans have been widely used as disposal sites for industrial waste. This was particularly true off the coast of southern California where waste streams associated with the oil and gas industry, the military, and the manufacturing of chemicals including DDT, were legally discharged into the sewer or dumped directly into the ocean from the 1930s to the early 1970s, with the bulk reportedly dumped in areas known as dumpsites 1 and 2 (1). Among these wastes, those contaminated with the organohalide insecticide DDT have garnered considerable attention

due to its persistence in the environment, bioaccumulation properties, and detrimental effects on human and animal health (2–5).

The chief source of DDT contamination off the coast of California was the Montrose Chemical Company, which produced ca. 800,000 tons of DDT between 1947 and 1982 (6). Montrose legally discharged thousands of gallons of DDT-contaminated acid and alkaline waste into the Los Angeles municipal sewer system, which heavily contaminated the Palos Verdes Shelf (1). Waste disposal was extended to offshore sites, with barges used to transport both bulk and containerized waste. The latter included steel barrels

Competing Interest: The authors declare no competing interests.

Received: February 14, 2025. **Accepted:** July 28, 2025

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that were reportedly punctured to facilitate sinking (1). It was estimated that >300,000 barrels were dumped in the San Pedro Basin (1), with a recent side-scan sonar survey identifying ca. 27,000 “barrel-like” objects on the seafloor (7).

Dramatic images of decomposing steel barrels on the San Pedro Basin seafloor and their linkage to DDT waste disposal have generated extensive media coverage and renewed concerns about the potential health hazards of DDT contamination. The first exploration of sediments surrounding these barrels revealed variable concentrations of DDT and its breakdown products (dichlorodiphenyldichloroethylene (DDE), dichlorodiphenyldichloroethane (DDD), and 1,1'-(2-chloro-1,1-ethenediyl)bis(4-chlorobenzene) (DDMU), collectively referred to as DDX), with the highest observed concentrations (257 µg/g sediment) exceeding those reported for shallow sediments from the nearby Palos Verdes Shelf superfund site (8). It was estimated that these barrels contained up to 1,535 tons of DDT and that they may account for the high levels of DDT and its associated breakdown products recorded in San Pedro Basin sediments (8). Given that DDX concentrations were not correlated with proximity to a barrel in a prior study, it was concluded that the contents escaped containment and entered the sedimentary system (8). In a more recent study, it was suggested that DDT waste was dumped in bulk and that the contents of the barrels may have included low-level radioactive waste (9). Given their poor condition and the more than five decades since these barrels settled on the seafloor, it would be expected that their contents have long since dispersed. As such, and given the poor historic record, it has not been possible to determine what these barrels contained.

Along with the discovery of the barrels, it was noted that some included concrete footings, described as intentional ballasting, and were surrounded by white, ring-like structures described as microbial mats (8, 10). An analysis of the microbial communities associated with the white rings surrounding two barrels, henceforth referred to as “white halo” sediment samples, revealed reduced diversity relative to the surrounding sediments, as is consistent with an environmental disturbance (11). These samples were dominated by the obligately anaerobic genus *Desulfobacula*, which is not known to form microbial mats (12), thus raising questions about the composition of the white ring structures. Given the challenges associated with accessing these deep-sea sites, ambiguities remain about the contents of the barrels, the composition of the white ring structures and solid concretions that surround some barrels, and the processes that led to their formation.

We initiated this study to address the effects of containerized waste contaminated with DDT on sediment microbial community composition and the potential role of microbes in DDT remediation. During the process of sampling sediments around corroding steel barrels on the San Pedro Basin seafloor, we observed two types of barrels, those surrounded by concretions and white halos and those that lacked these features. While we found no evidence linking either barrel type to DDT contamination, we did find evidence that the detection of a white halo and concretion around a barrel is a visual predictor that the barrel contained alkaline waste. We further ascribe the formation of the concretions and white halos to magnesium hydroxide and calcium carbonate precipitation, respectively, both anthropogenically induced from alkaline waste leakage. Remarkably, containerized alkaline waste discarded >50 years ago represents a persistent pollutant that has created extremophile hotspots resembling those observed around some hydrothermal fields and other natural ecosystems. The precise number of these hotspots and their collective effect on the benthic ecology of the San Pedro Basin remains to be

determined, although their effects on macrofaunal assemblages have been documented (10).

Results and discussion

Sample collection

Sediment push cores were collected as previously described (10) during two remotely operated vehicle (ROV) dives to an average depth of 901 m within an area of the San Pedro Basin (dumpsite 2) known to be contaminated with DDT (8). The primary objective was to collect sediment samples around corroding steel barrels linked to the dumping of industrial waste in this nearly anoxic basin (average dissolved oxygen concentration 0.45 µmol/L). The two dive sites differed in the number of barrels observed (Fig. 1) and their visual appearance (Fig. 2). Only two barrels were observed over the roughly 2-km-long S0450 dive track, and sediment cores were collected along transects adjacent to both (barrels 1N and 2N). These barrels were highly degraded and neither had a white halo surrounding them (Fig. 2D). In contrast, 22 relatively intact but still structurally compromised barrels were observed over the considerably shorter (ca. 400 m) S0451 dive track. Six of these barrels were surrounded partially or completely by a white halo, and sediment cores were collected along transects perpendicular to three of these barrels (barrels 1S–3S, Figs. 1 and 2C and E). Coordinates, photographs, and descriptions of the barrels, both sampled and observed, are provided in Table S1 and File S1 with additional metadata reported elsewhere (10). The patchy distribution of the barrels is concordant with barged disposal (1, 9) and further evidenced in large-scale surveys of the area (7).

Sediments were collected using push cores placed 5, 3, 1 m (through the white halo if present), and adjacent to each barrel (0 m) when possible (Fig. 2A and B). When a white halo was present, the sediment between the halo and the barrel could not be penetrated due to the presence of hard substrate usually covered by a thin layer of sediment. This hard substrate was previously postulated to represent concrete ballasting (8) or brucite formed when leaked barrel contents interacted with the sediment (7, 10), although no evidence has been presented to support either hypothesis. A sample of the concretion surrounding one barrel was collected using the ROV arm. Control cores at sites without visible barrels were collected during both dives (two on dive S0450 and one on dive S0451) and from four areas outside dumpsite 2 with similar depth and oxygen regimes (Fig. 1). In total, 26 sediment cores were collected and sectioned into 4 × 2 cm depth horizons of which 97 horizons were successfully processed for microbial community and DDX analyses.

Sediment DDX concentrations

DDX concentrations were recorded for all 97 sediment samples, with the highest values consistently found 2–6 cm below the sediment surface (Table S2). This is largely consistent with prior reports (8–10), as was the lack of correlation between DDX concentration and distance to a barrel (Fig. S1). The DDX values detected for the three dumpsite 2 control cores were also largely consistent with those observed around the barrels (Table S2), which suggests that the contents of the barrels either were not contaminated with DDT or were broadly dispersed after disposal. Nonetheless, the DDX concentrations observed across all dumpsite sediments averaged 233.8 ± 326.6 ng/g dry weight of sediment (ng/g), with a high of 1,404 ng/g dry weight. This average is two orders of magnitude greater than the 3.0 ± 2.0 ng/g average observed for sediments collected from the four sites outside of

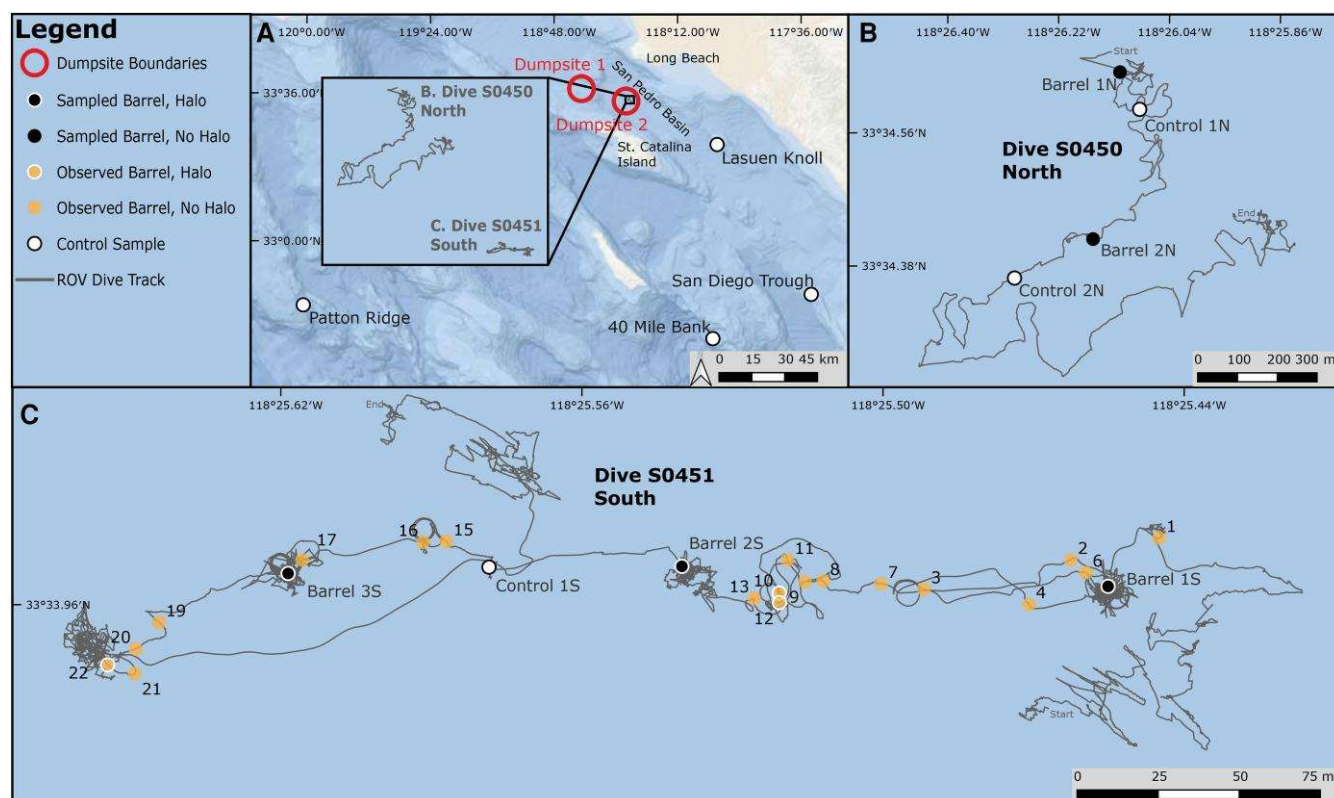


Fig. 1. ROV dive tracks and sediment sampling. A) Location of two reported dumpsites (red circles) in the San Pedro Basin. Inset: ROV tracks for dives S0450 and S0451. B) Detail of dive S0450 indicating barrels sampled (black circles) and control cores (white circles). C) Detail of dive S0451 indicating barrels sampled (black circles), barrels observed but not sampled (orange circles), barrels with white halos (white outline around circle), and control cores (white circles). Numbers indicate barrel ID (Table S1). Map image is the intellectual property of Esri and is used herein under license. Copyright © 2025 Esri and its licensors. All rights reserved.

dumpsite 2, where the maximum detected was 8.8 ng/g (Table S2). This provides evidence that the high levels of DDX contamination observed within the dumpsite have not extended as far southeast (43 km) as Lasuen Knoll (Fig. 1).

Microbial community analyses

Given the varying but generally high levels of DDX contamination detected in San Pedro Basin dumpsite 2 sediments, we asked whether this pollutant influenced microbial community composition by generating 16S rRNA gene amplicon libraries for all 97 sediment samples. The resulting sequences were taxonomically diverse and assigned to 79,112 bacterial and archaeal amplified sequence variants (ASVs). After clustering the ASVs into 40,212 operational taxonomic units (OTUs) at 97% sequence identity (ID), there was no significant (i.e. $P < 0.05$) relationship between DDX concentration and community composition (Bray–Curtis permutational multivariate analysis of variance, PERMANOVA) when considering all San Pedro Basin samples (Table S3) or when restricting the analyses to the depth horizons with the highest DDX concentrations (2–4 and 4–6 cm, total DDX pseudo $F = 0.79$, $R^2 = 2.3\%$, $P = 0.6$). In addition, we found no significant correlation between DDX concentration and Shannon's diversity index (13) or Faith's phylogenetic diversity (14) (Fig. S1). In contrast, sediment depth horizon and sampling location impacted community composition to a larger degree than DDX concentration (Table S3 and Fig. S1). While these results suggest that DDX contamination is not a major driver of sediment microbial community structure, we continued to examine how the contents of the barrels may have affected community composition.

We next assessed the sediment cores collected from the white halos to explore prior proposals that these structures were microbial in origin. Obtaining DNA from the halo sediments was challenging and required repeated extraction, additional clean-up steps, and the pooling of several extracts to obtain low yields (average 1.0 ± 28.5 ng/ μ L) of poor-quality DNA (Nanodrop 260/280 average 2.80; 260/230 average 0.62). Rarefaction curves (Fig. S2) and Faith's phylogenetic diversity metrics (Fig. 3) revealed that the halo sediment communities were significantly less diverse (average Faith's PD = 64.4 ± 39.2) than other samples (average Faith's PD = 151 ± 44.2 , Kruskal–Wallis $P = 0.0015$) (Table S4) and contained fewer OTUs (Table S2), averaging 428 ± 156 (barrel 1S, 6–8 cm outlier removed) compared with $1,512 \pm 592$ for the nonhalo sediments. Furthermore, the 10 most abundant OTUs in the white halo sediments comprised on average $63.4\% \pm 17.6\%$ of the community compared with $18.7\% \pm 3.5\%$ in the nonhalo sediments (Fig. 4). Similarly, reduced macrofaunal density and diversity have been reported in sediments underlying the white halos (10).

In contrast to the white halo sediments, the microbial diversity metrics for the nonhalo sediments (richness, Shannon's diversity index, Faith's phylogenetic diversity, and relative abundances) were not significantly different across sediment depth horizons or distance from the barrels (Kruskal–Wallis $P > 0.1$). These communities were highly diverse, evenly distributed, and shared similar fingerprints, as previously shown for San Pedro Basin sediments (15, 16). The most abundant of the 82 bacterial phyla detected were Thermodesulfobacteriota (22%), Planctomycetota (17%), Chloroflexota (8.6%), Pseudomonadota (previously Proteobacteria) (8.5%), Bacteroidota (4.8%), Acidobacteriota (4%),

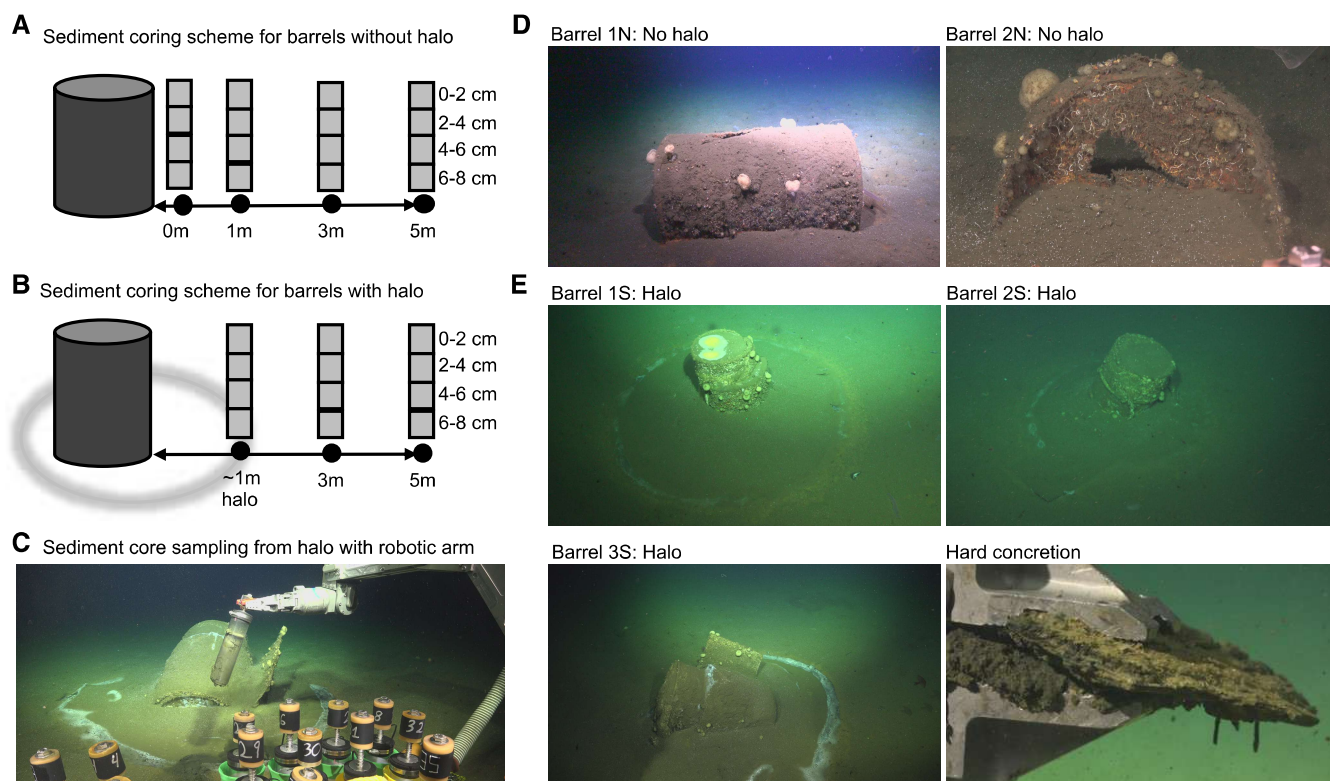


Fig. 2. Barrel sampling. Barrels without (A) and with a white halo (B) were sampled as shown. Sediment cores are depicted as gray columns divided into four 2-cm depth horizons. C) ROV recovering a sediment push core. D) Two barrels without white halos (1N and 2N) were sampled during dive S0450. E) Three barrels with white halos (1S–3S) were sampled during dive S0451 along with the concretion surrounding barrel 3S.

and Nitrospirota (2.6%), while the dominant archaeal phyla included Nanobdellota (previously Nanoarchaeota) (4.8%) and Thermoprotei (previously Crenarchaeota) (2.6%), with archaea accounting for ca. 11% of the total microbial community. The non-halo dumpsite 2 sediment communities were nonetheless distinct from those detected outside the dumpsite (Fig. S3 and Table S4), suggesting that pollution levels in this area are a contributing factor.

The dramatic shift in the microbial communities associated with the white halo sediments extended throughout the core for barrel 2S, to the 4–6-cm horizon for barrel 1S, and in the 0–2-cm horizon for barrel 3S, which was the only sample from that core where DNA was successfully obtained (Fig. 4). The reduced diversity in these sediments was linked to the high relative abundance of a few gram-positive OTUs in the phylum Bacillota (Fig. 4) that are closely related to bacteria capable of growth in alkaline conditions. Collectively, the 10 most abundant OTUs in these sediments (outlier removed, Tables S5 and S6) accounted for 71% of the microbial communities, a remarkable drop in diversity compared with what is traditionally observed in marine sediments (17). Among these, OTU 1 accounted for up to 55% of the relative abundance. A closely related strain is *Serpentinicella alkaliphila* (NR 152685, 94.4% ID), which grows at pH 10.1 and was isolated from a carbonaceous, hyperalkaline spring (Fig. 5 and Table S5) (18). The closest cultivated relative of OTU 2 (up to 42% relative abundance) is *Ca. Desulfuridis audaxviator* (CP034260, 96.8% ID), isolated from a 2-km-deep aquifer (19) and detected globally in deep-subsurface fluids (mgm48515403, mgm48515433, 96.8% ID) (20). These fluids are often alkaline (21, 22) and inhabited by hyperalkaliphilic microorganisms such as *Alkaliphilus transvaalensis* (AB037677), which grows at pH 12.5 (23). OTU 3 (up to 34% relative abundance) was closely related to *Alkalicella caledoniensis* (NR 173660,

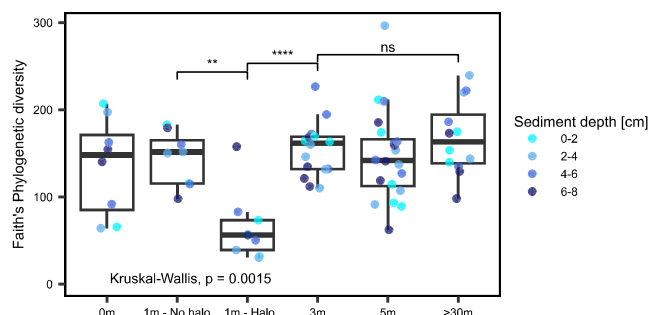
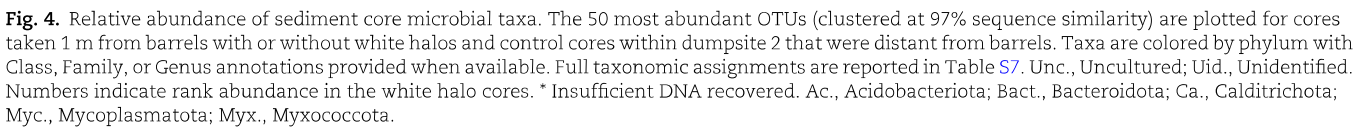


Fig. 3. Sediment microbial diversity. Box and whisker plot with overlaid jitter plot of Faith's phylogenetic diversity in sediment samples collected at different distances from barrels (X-axis). Horizontal line indicates the median, box shows the interquartile range, and whiskers extend to $\pm 1.5\times$ interquartile range. Colors correspond to sediment depth horizons. *Indicates significant difference between means; ns indicates not significant.

98.8% ID), which grows at pH 10.8 and was isolated from a pH 9.0 alkaline thermal spring (24). Other OTUs of interest include OTU 6 (up to 11% relative abundance), which is closely related to *Peloplasma aerotolerans* (OR436924, 97.2% ID), an anaerobic, free-living mollicute isolated from a pH 8.5 mud volcano (25). Mollicutes are poorly studied and have also been detected in hyperalkaline springs in California (26). Canonical mat-forming bacteria such as sulfur-oxidizing Thiotrichales accounted for maximally 0.15% relative abundance (barrel 2S, 0–2 cm, 40 reads), suggesting that they are not associated with the formation of the white halo structures.

A phylogeny of the dominant white halo OTUs and their top environmental (uncultured) BLAST matches supports their close



Sediment pore-water pH

and similarly large differences between the halo samples and the surrounding sediments (Fig. S4). While the results from the different measurement types were positively correlated ($R^2=0.92$, $P<0.001$), how well these measurements reflect *in situ* conditions remains unknown. Nonetheless, the pore-water pH associated with the white halo sediments is clearly elevated by several units, while the values recorded for the nonhalo sediments were in concordance with prior reports for southern California Borderland basin sediments (28).

The pore-water pH values and microbial community analyses provide strong evidence that the sediments underlying the white halos are highly alkaline. By extension, the observation of a white halo around a barrel is a strong predictor that the barrel contained alkaline waste. Given that six of the 24 barrels observed during ROV dives S0450 and S0451 (File S1) were partially or completely surrounded by a white halo (Fig. 1 and Table S1), we can estimate that 25% contained alkaline waste. To extrapolate further, we analyzed all publicly available images from ROV dives in the area (8) and identified partial or complete white halos surrounding 15 of 37 additional barrels (41%). When compiled, we estimate that 21 of the 61 visually identified dumpsite 2 barrels (34%) contained alkaline waste. Given that more than 74,117 debris targets of unknown composition have been reported from dumpsite 2 (7), the aggregate ecological impacts of containerized alkaline waste disposal could be highly significant.

Given the elevated pore-water pH recorded for the white halo sediment samples, we next examined whether this was a major driver of community structure. In general, the levels of elevated pore-water pH were largely consistent with the shifts observed in community composition (Fig. 4). For example, the pH levels of the 0–2-cm horizons for barrels 1S and 2S were lower than those recorded for the deeper horizons, possibly due to dilution with the overlying seawater. Correspondingly, the relative abundance of Bacillota was also reduced in these sediments. Similarly, the extremely high pore-water pH (11.5) recorded for the 0–2-cm horizon of barrel 3S corresponded to sediment communities highly enriched in Bacillota. The one exception was the 6–8-cm horizon from barrel 1S, where high pore-water pH was recorded along with a diverse community that resembled the nonhalo sediments (Fig. 4). This outlier could represent the transition zone between high and neutral pH horizons where our processing (thorough

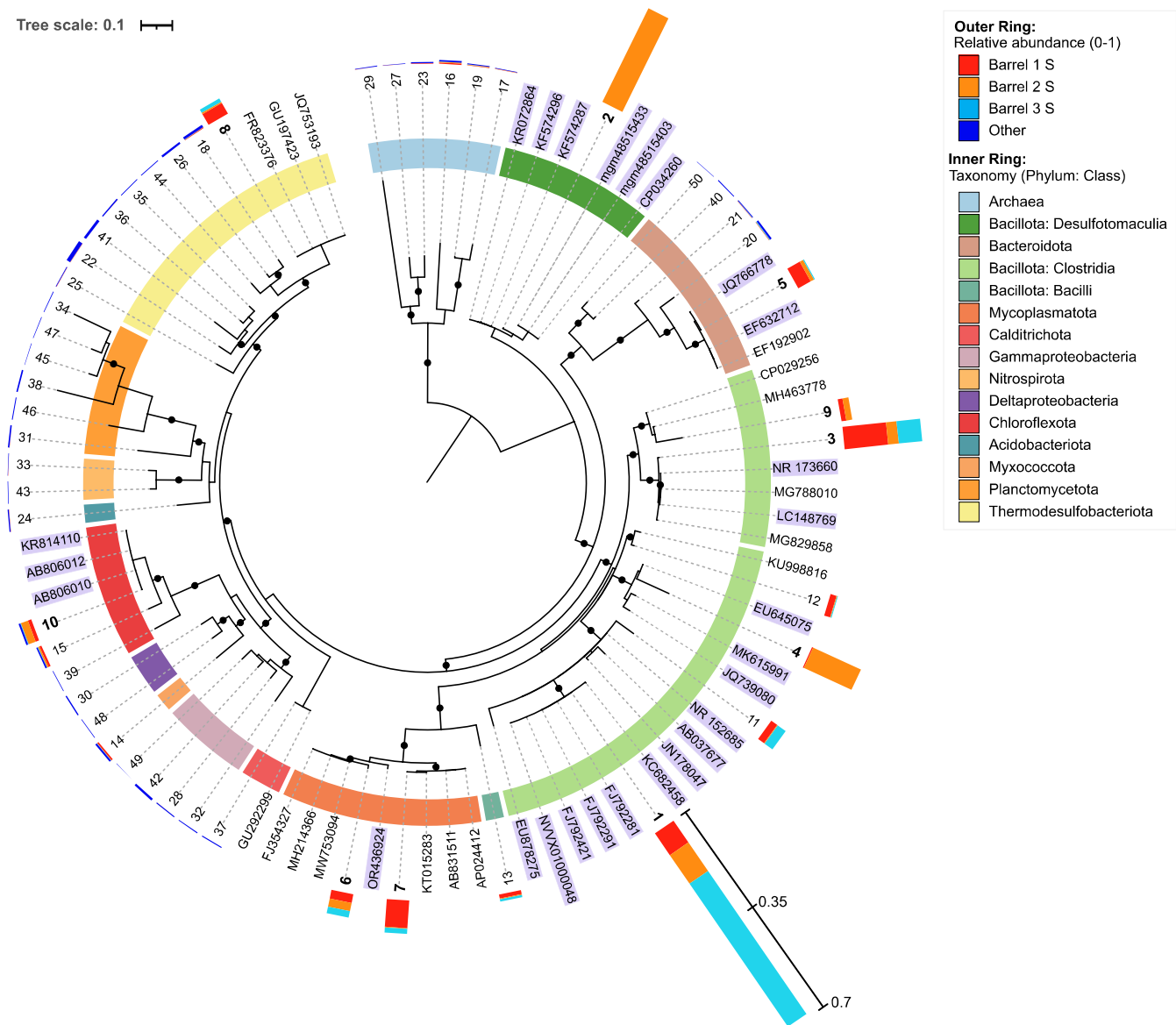


Fig. 5. Maximum likelihood phylogeny and relative abundance of sediment taxa. FastTree includes the 50 most abundant OTUs (clustered at 97% sequence similarity, numbered by rank abundance, and delineated by taxonomy, inner circle) detected in sediment cores collected 1 m from barrels either with a white halo (barrels 1S–3S) or without a halo (Other: 1 m from nonhalo barrels plus dumpsite 2 controls as in Fig. 4). OTU counts were summed per core for white halo cores, and across all included no-halo cores for “Other” before calculating relative abundances. Top BLAST matches and other closely related taxa to OTUs 1–10 (bold) were included in the tree (NCBI or MG-RAST accession numbers given, details listed in Table S5) and colored lavender if the isolation/detection source was an alkaline environment. Archaeal OTUs were used as outgroups. Bootstraps >80% are indicated by black circles.

mixing) of the 2-cm section, coupled with the logarithmic nature of pH, resulted in an elevated pore-water pH measurement yet a microbial community that resembled neutral conditions. To further explore these observations, a nonmetric multidimensional scaling (NMDS) plot of Bray–Curtis dissimilarities (Fig. 6A) shows the halo sediment communities as highly divergent from all other samples with pH being the highest correlated environmental factor ($R^2=0.57$) explaining variability in the first dimension (vegan function envfit, 999 permutations, $P=0.001$). Among nonhalo samples, sediment depth ($R^2=0.24$) explained some variation along the second dimension ($P=0.001$) and was slightly cocrrelated with total DDX concentrations (Pearson’s $r=-0.12$, $P<0.01$).

Comparison of microbial communities with other hyperalkaline environments

The dumping of containerized alkaline waste appears to have created environmental conditions on the seafloor that mimic naturally occurring, hyperalkaline environments such as those reported from serpentinizing hydrothermal fields and hyperalkaline springs (29, 30). Given this, we compared the structural and phylogenetic relatedness of microbial communities from the San Pedro Basin to those associated with the Lost City hydrothermal field, pH 9–11 (29) and serpentinizing, hyperalkaline springs in the Italian Voltri Massif, pH 8–12 (30), which are the only comparable alkaline environments we could identify on the Qiita platform (31). A weighted UniFrac analysis revealed that the San Pedro Basin white halo samples clustered more closely with several Voltri Massif

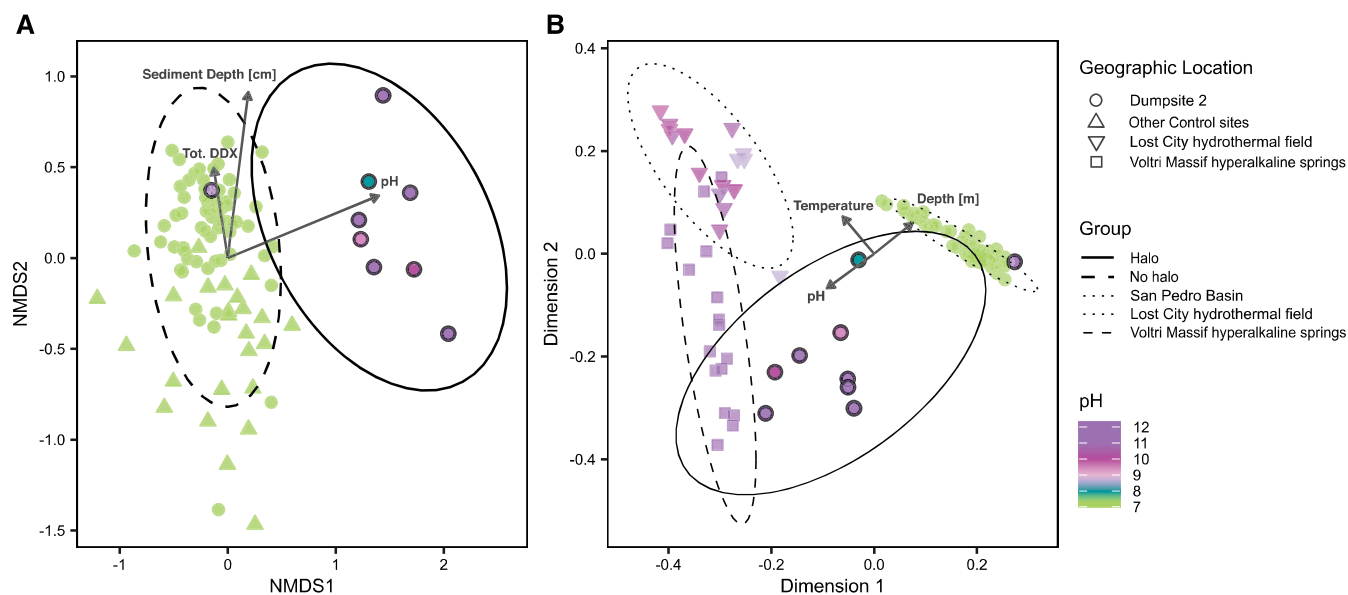


Fig. 6. Multivariate comparisons of microbial communities from dumpsite 2 (circles) with other sites. Colors correspond to pore-water pH estimates. Black outlines indicate samples from halos. Ellipses indicate 95% clustering confidence intervals for groups of samples. Arrows represent fitted environmental factors explaining variation in these directions (scaled by correlation coefficient). A) NMDS plot of microbial community Bray-Curtis dissimilarities from dumpsite 2 (circles) compared with other control sites (triangles). Tot. DDX, total DDX concentration (ng/g). B) Weighted UniFrac PCoA of microbial communities from dumpsite 2 (circles) compared with other hyperalkaline environments. White halo communities (black outlines) cluster with Voltri Massif hyperalkaline springs samples (squares) and more distantly with samples from the Lost City hydrothermal field (inverted triangles).

hyperalkaline spring (VMHS) samples than to physically adjacent sediments from the San Pedro Basin (Fig. 6B). After geographic location (adonis $R^2 = 0.479$, $F = 73.08$, $P = 0.001$), pH explained most of the variation in community composition (adonis $R^2 = 0.093$, $F = 28.49$, $P = 0.001$, 999 permutations) and was the environmental factor driving separation in this direction (Fig. 6B). Like the white halo sediment communities in the San Pedro Basin, *Clostridia* were abundant in the VMHS communities and also other serpentizing sites (29, 32–34). Other shared taxa include free-living Mollicutes, Bacteroidetes, and anaerobic sulfate-reducing Thermodesulfobacteriota (Desulfovibrionales) detected in the VMHS or other hyperalkaline springs (26, 29, 33, 35).

Metagenomic analysis

To better understand the genetic basis for microbial adaptation to the hyperalkaline white halo sediments, we generated metagenomes from all four depth horizons obtained from two white halo cores and one nonhalo control core (Table S8). One horizon from one of the halo cores (barrel 2S, 6–8 cm, SCB-300b-D) yielded insufficient reads and was excluded from the analyses. Each metagenome was assembled independently, and all predicted proteins from the resulting contigs annotated using Prokka (36). One-sided homoscedastic models of these annotations revealed that the halo-derived metagenomes contained significantly higher frequencies of multisubunit Mrp-type Na^+/H^+ antiporters than control (no-halo) samples (Fig. 7 and Table S9). These antiporters provide a well-established mechanism to maintain cytoplasm pH homeostasis under extreme alkaline conditions (37, 38). The white halo sediment metagenomes were also significantly enriched in sodium-driven ATPases and sodium-coupled symporters for organic solute uptake, both of which are associated with overcoming a reduction in proton motive force at high pH (39, 40).

Metagenome-assembled genomes (MAGs) were reconstructed by binning assembled contigs from each white halo sample and sediment horizon. Five nearly complete MAGs closely related to

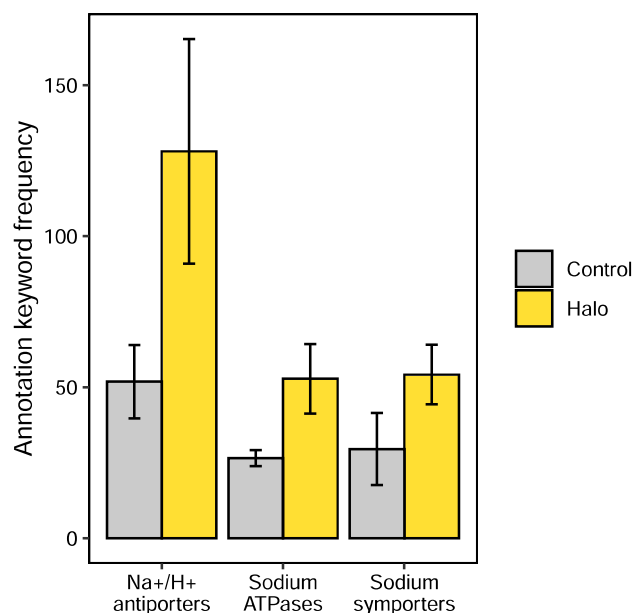


Fig. 7. Protein functional keyword annotation frequencies for alkaliphilic characteristics in metagenomic assemblies. Bars show averages and SDs (normalized per 100,000 predicted proteins) for white halo and nonhalo control sediments. Data reported in Table S9.

the dominant Natronincolaceae lineage observed in the microbial community analyses (OTU 1 in Figs. 4 and 5) were found in both halo cores (Table S10). These MAGs were not assembled from the nonhalo control site metagenomes, where the corresponding Natronincolaceae 16S rRNA gene sequences were also not observed (Fig. 4). Based on average amino acid identities (AAIs) of >99.5%, these five nearly identical MAGs all belong to the same species (Fig. S5). The closest sequenced MAG, with AAI scores of 82–83%, is *Alkaliphilus* sp. NORP52 (Clostridiaceae), recovered

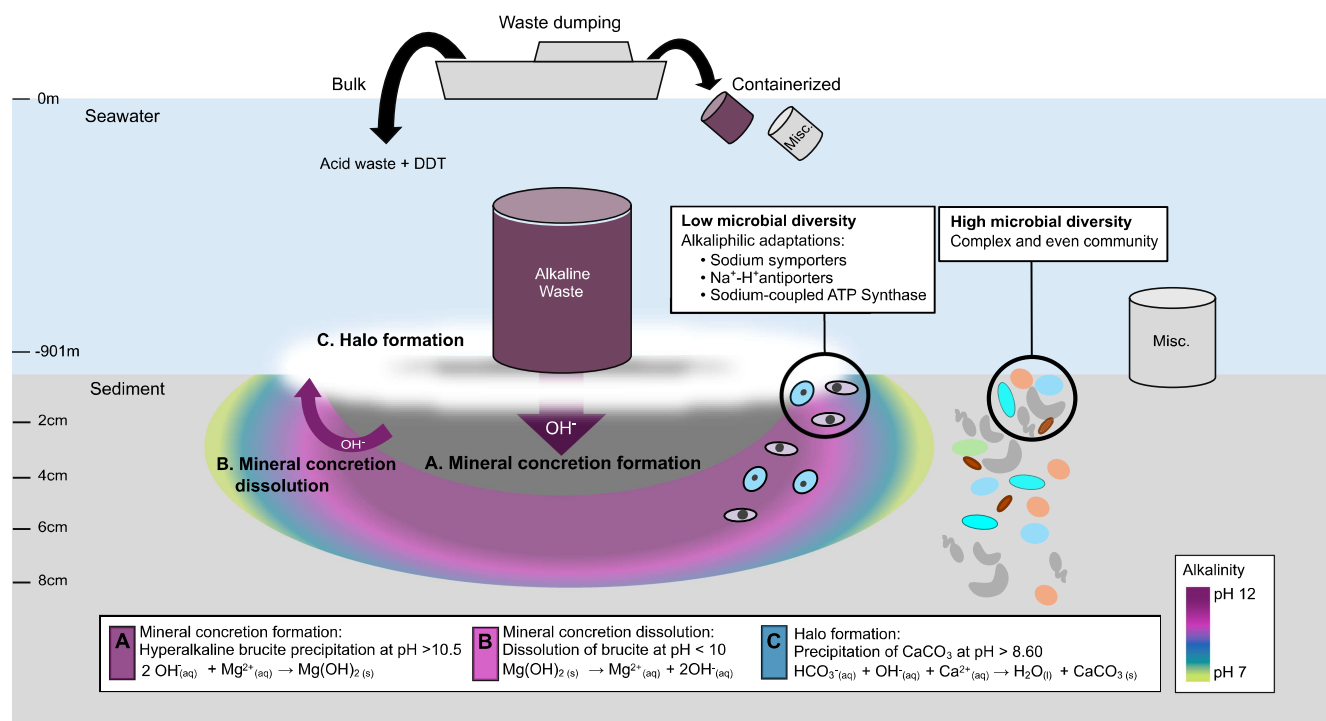


Fig. 8. Environmental impacts of alkaline waste disposal. A) Alkaline waste (OH^-) leaked from a steel barrel after settling on the seafloor interacts with magnesium ions in sediment pore waters inducing the formation of a brucite ($\text{Mg}(\text{OH})_2$)-enriched concretion extending ca. 1 m out from the barrel. B) Over time, a gradual reduction in pore-water pH induces brucite dissolution, and either alone or in combination with residual alkaline waste, triggers the precipitation of calcium carbonate around the rim of the concretion (C) as the alkaline pore water (pH 8.6–10) fluxes up through the sediment. Extreme alkalinity selects for a low diversity and highly adapted microbial community in sediments below the white halos, while communities outside the halos remain highly diverse.

from subsurface crustal aquifers on the Mid-Atlantic Ridge (41). This level of sequence identity places the five white halo MAGs in the genus *Alkaliphilus* (42). The next most closely related genomes were all Natronincolaceae family members (Fig. S5 and Table S10) isolated from highly alkaline environments, including marine serpentinizing hydrothermal fields (*Alkaliphilus pronyensis*, *A. serpentinus*, *A. hydrothermalis*, and *Serpentinicella alkaliphila*), the deep subsurface (*A. transvaalensis*), and soda lakes (*Alkaliphilus multivorans*, *Natronicola peptidivorans*, *Natronicola ferrireducens*, and *Serpentinicella* sp. ANB-PHB4). Genetic evidence for a highly specialized alkaliphilic lifestyle among the five *Alkaliphilus* MAGs includes two full copies of the Mrp antiporter operon (Table S10), while most other Natronincolaceae genomes contain only one. Further evidence includes the ATP synthases, which were exclusively sodium-driven. In contrast, previously described facultative alkaliphiles in the Natronincolaceae family encode both sodium- and proton-driven synthases (Table S10).

Mineral analysis

Our microbial community analyses indicate that the white halo sediments surrounding some barrels are not dominated by mat-forming, sulfur-metabolizing bacteria as previously suggested (8). Instead, we propose that they represent mineral precipitates induced by environmental conditions. To further examine this, we compared the mineral content in the 0–2-cm horizons between white halo ($n=3$) and nonhalo ($n=6$) cores using X-ray diffraction (43). We found differences in the two most abundant minerals, with calcite more abundant than mica in most halo sediments and mica more abundant than calcite in most no-halo sediments (Table S11). Notably, brucite is a major mineral component of halo sample 310bA, which also contains more calcite than mica. The

other minerals measured (quartz, plagioclase, halite, amphibole, chlorite, and sepiolite) showed similar abundance between the white halo and nonhalo samples. We suggest that these mineralogy results correspond to increasing alkalinity driving calcite (44) and some brucite precipitation concomitant with mica dissolution (45) in the white halo sediments. We further postulate that the white halo materials consist of calcium carbonates that have precipitated from seawater in response to contact with alkaline fluids diffusing upward through the sediments, with mineral concretions defining the area around the barrels where diffusion and precipitation occur.

We next addressed the composition of the concretion surrounding barrel 3S and determined that it was a mixture of predominantly brucite as well as calcite, aragonite, mica, and quartz (Fig. S6 and Table S11). While calcium carbonates, clay, and quartz are common constituents of seafloor sediments, brucite is not, with exceptions including hydrothermal vent systems such as the Lost City hydrothermal field, where it forms through the process of serpentinization (46). This geochemical process leads to the formation of minerals such as serpentinite, brucite, and magnetite while releasing hyperalkaline water that can surpass pH 12 (47–50). Upon contact with seawater, these hyperalkaline fluids precipitate calcium carbonates, forming large, strikingly white, calcite–brucite chimneys at active venting sites (51, 52). As observed in the white halo sediments, these alkaline serpentinization environments are associated with low microbial abundances and diversity that is largely limited to anaerobic or microaerophilic microorganisms adapted to high pH (48, 53). Apart from the concretion, brucite was absent in all sediments tested except in the white halo core from barrel 3S, which was adjacent to the crust and for which brucite was determined to be a major mineral component (Table S11).

Impacts of alkaline waste disposal

We postulate that alkaline waste contained in steel barrels and dumped into the sea off Los Angeles settled to the seafloor and leaked into the surrounding sediment, thus triggering magnesium precipitation into brucite ($\text{Mg}(\text{OH})_2$)-enriched concretions surrounding the barrels (Fig. 8). Subsequent brucite dissolution, which increases sediment pH by releasing hydroxides (54), either alone or aided by residual alkaline waste, triggers the precipitation of carbonate minerals from the overlying seawater. This white precipitate forms a halo on the sediment surface around the rim of the concretion. The effects of these anthropogenically induced processes bear remarkable resemblance, in terms of both mineralization and changes to the sediment microbial communities, to those observed in naturally occurring hyperalkaline environments. The reduced microbial abundances and diversity associated with these extremophile hotspots have unforeseen consequences for ecosystem function and services. Given rate estimates for exposed brucite dissolution in seawater (46, 54, 55), it could take several thousand years for the effects of caustic alkaline waste dumping in the San Pedro Basin to be resolved.

Summary and conclusions

Marine microbial communities mediate essential ecological processes that are critical to the health of the oceans and planet. The introduction of anthropogenic pollutants can alter microbial community structure and function with generally unknown long-term effects. This is especially true of the deep sea, where few studies have documented the effects of extensive waste dumping on microbial community composition. Between 1961 and 1964, the Pacific Ocean Disposal Company dumped 1,382,000 gallons of alkaline waste in the San Pedro Basin (1). Reports do not clearly indicate what, if any, of this waste was containerized, and much speculation has linked corroding steel barrels observed on the San Pedro Basin seafloor to the high levels of DDT contamination reported in the area (8). Our results provide evidence that some of these barrels contained alkaline waste. The leakage of this waste is associated with altered microbial communities adapted to hyperalkaline conditions, the formation of mineral concretions, and white rings of calcium carbonate precipitate surrounding the barrels. These visual cues provide a rapid method to predict which barrels contained alkaline waste, which account for ca. one-third of the steel barrels visually identified to date on the San Pedro Basin seafloor. While persistent organic pollutants such as DDT have been the primary concern for the San Pedro Basin dumpsites, the localized effects of containerized alkaline waste dumping persist more than 50 years after disposal and thus represent an unanticipated, persistent pollutant of concern.

Methods

See [SI Methods](#).

Acknowledgments

The authors acknowledge the analytical and coordinating team at Physis Environmental Laboratories, especially Rich Gossett and Mark Baker, for analyses of DDX concentrations in sediments. In addition, the authors acknowledge the Schmidt Ocean Institute for providing ship time on R/V Falkor and the crews of the R/V Falkor, ROV SuBastian, R/V Nautilus, ROV Hercules, and R/V Robert Gordon Sproul who facilitated sample collection. We

acknowledge the Schmidt Ocean Institute for the unaltered pictures of the barrels shared under Creative Commons License CC BY-NC-SA 4.0. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the US government.

Supplementary Material

Supplementary material is available at [PNAS Nexus](#) online.

Funding

This research was funded by the National Oceanographic and Atmospheric Administration award nos. NA23NMF4690462 and NA22OAR4690679 to P.R.J. and L.A.L. and the University of Southern California Sea Grant award SCON-00003146 to L.A.L.

Author Contributions

Johanna Gutleben (Data curation, Formal analysis, Investigation, Methodology, Writing—original draft), Sheila Podell (Formal analysis, Investigation, Methodology, Writing—review & editing), Kira Mizell (Formal analysis, Methodology, Writing—review & editing), Douglas Sweeney (Formal analysis, Methodology, Writing—review & editing), Carlos Neira (Formal analysis, Methodology, Writing—review & editing), Lisa A. Levin (Funding acquisition, Methodology, Writing—review & editing), and Paul R. Jensen (Conceptualization, Funding acquisition, Methodology, Supervision, Writing—review & editing)

Data Availability

All data associated with this study are available to the public. The MAGs and 16S rRNA gene sequence data generated for this study have been deposited at DDBJ/ENA/GenBank under BioProject accession PRJNA1200136. Videos of sampling and all observations are available on YouTube on the Schmidt Ocean Channel (@SchmidtOcean) under the titles “ROV Dive 450—DDT Barrel Site 1 (North)” and “ROV Dive 451—DDT Barrel Site 2 (North).” X-ray diffraction scan data are published and available in the U.S. Geological Survey ScienceBase data release (43).

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